

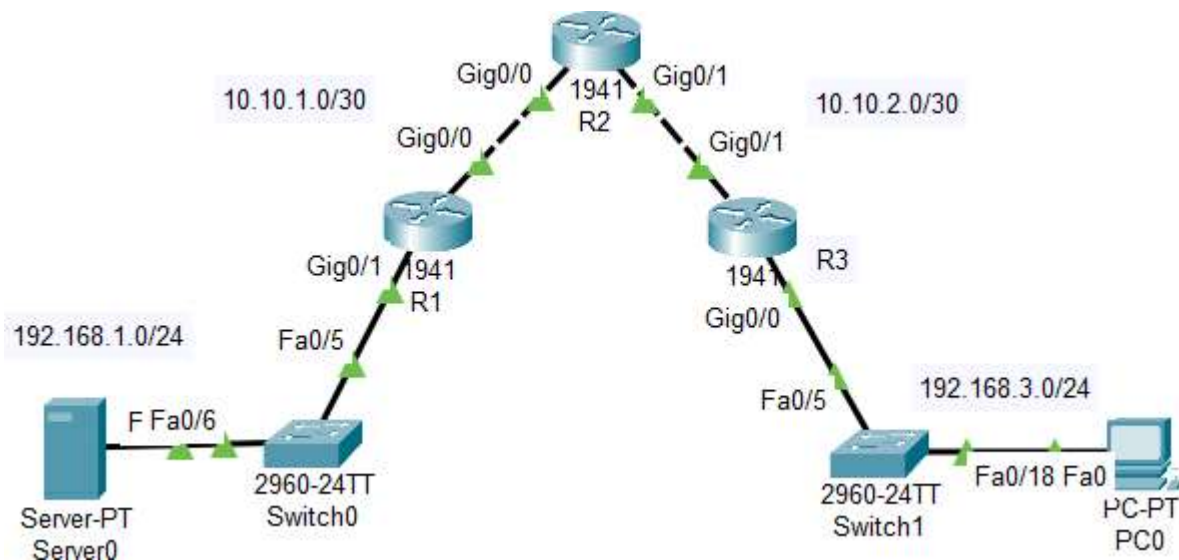
CCS6224

NETWORK SECURITY

Lab 3

Practical Lab: Security Configuration of Router for Administrative Access using AAA and Radius

1. Open your saved packet tracer configuration file in Lab 2. The following network topology is shown.



2. Secure IOS (Internetwork Operating System) image and configuration files on router R1.

- Check the IOS file in the router's flash storage
- Secure the IOS image
- Backup a copy of the running configuration file by archiving it in flash storage
- Verify that the IOS image and configuration are secured, what are the names for the two archived files?
- Check the available files in the router's flash storage again, are the two files in the list?
- Un-archive the IOS image and configuration file

3. Configure a Synchronized Time Source Using Network Time Protocol (NTP). R2 is the master NTP server in this lab. All other network devices learn the time from it, either directly or indirectly.

- a) Display the current time at R2. Then set the time at the router by using the command `clock set`
- b) Configure NTP authentication by defining the authentication key number, hashing type (md5), and password = NTPcisco
- c) Configure the trusted key that will be used for authentication on R2. Enable the NTP authentication on R2
- d) Configure R2 as the NTP master. Configure R1 and R3 as NTP clients

4. Configure syslog Support on R1 and Server-0.

- a) Turn ON the syslog service on Server-0
- b) Enable the timestamp service for logging on the router. You can verify whether the timestamp service for logging is enabled or not on the router using the `show run` command
- c) Configure the syslog service on the router to send syslog messages to the syslog server
- d) Configure the logging severity level on R1

Routers support different levels of logging. The eight levels range from 0 (emergencies), indicating that the system is unstable, to 7 (debugging), which sends messages that include router information.

alerts	Immediate action needed	(severity=1)
critical	Critical conditions	(severity=2)
debugging	Debugging messages	(severity=7)
emergencies	System is unusable	(severity=0)
errors	Error conditions	(severity=3)
informational	Informational messages	(severity=6)
notifications	Normal but significant conditions	(severity=5)
warnings	Warning conditions	(severity=4)

- e) What will be the problem(s) if we set the level of severity too high? or too low?
- f) Display the current status logging for R1. What port is syslog using? What is the IP address for the syslog server?

5. Configure Local AAA on R3

- a) Create a local user named Admin03 with highest privilege, set password = Admin03pass
- b) Enable AAA services
- c) Create the default login authentication list for console access
- d) Exit from the console, and then login back to the console using the Admin03 user and its password, were you able to login?
- e) Create a AAA authentication profile for Telnet using the local database
- f) Open a Telnet session from PC0 to R3 using the created AAA authentication profile. Were you able to login?

6. Observe AAA Authentication Using Cisco IOS Debug

- First, verify that the time-stamp information is available for your debug output using the `show run` command. If not, enter `service timestamps debug datetime msec` command in the global configuration mode
- Activate debugging for AAA authentication
- Open a Telnet session from PC0 to R3, observe the successful and unsuccessful authentication attempts

7. Configure Centralized AAA Authentication on R2 Using RADIUS

- Configure enable secret = cisco123
- Enable AAA services on R2
- Create a local user named Admin04 with highest privilege, set password = Admin04pass
- Create the AAA authentication list using the Radius Server
- Set the IP address and key (tcs2351) of the Radius Server on R2
- Assign the authentication using the Radius Server on the Telnet line
- Save the configuration on R2
- Next, we need to setup the Radius Server at Server0, click on the service tab at Server0 and the following screen will appear

The screenshot shows the configuration interface for Server0, specifically the 'Services' tab. The 'AAA' service is selected in the left-hand menu. The main configuration area is titled 'AAA' and includes the following sections:

- Service:** A radio button for 'On' is selected, and the 'Radius Port' is set to '1645'.
- Network Configuration:** This section contains fields for 'Client Name', 'Client IP', 'Secret', and 'ServerType' (set to 'Radius'). Below these fields is a table with columns for 'Client Name', 'Client IP', 'Server Type', and 'Key', with 'Add', 'Save', and 'Remove' buttons to the right.
- User Setup:** This section contains fields for 'Username' and 'Password'. Below these fields is a table with columns for 'Username' and 'Password', with 'Add', 'Save', and 'Remove' buttons to the right.

- i) Turn ON the AAA service in Server0
- j) Configure the client details as the followings, and then add the client
 - Client name: R2
 - Client IP: 192.168.1.1
 - Secret (key): tcs2351
 - Server type: radius
- k) Create and add a username with password
- l) Open a Telnet session from PC0 to R2 using the centralized AAA authentication with the Radius Server. Were you able to login?
- m) Close the Telnet session
- n) Shutdown the Radius Server, Open another Telnet session from PC0 to R2. Were you able to login?

8. Save all your configuration in the pkt file.